CS1002 –Programming

Fundamentals

Assignment # 02

Instructor	**Muhammad Yousaf**
Session	**Fall 2023**
Section	**BCS 1D**

General Guidelines

1. Peer plagiarism and late submissions are strictly not allowed. In case, zero marks will be awarded for whole assignment.
2. Total Marks: 120

Submission Guidelines

1. You will upload the assignment on CLASSROOM in the given timeline.
2. Don't email your solution to the instructor for submission. Submit your assignment in the given deadline, said LMS.
3. No submission will be accepted later than said deadline.

Deadline: Oct 30, 2023, 11:00 PM

Q.No.1:

Write a program that will take input and display the Character Class and the class code for which the character belongs to. You can think of it as we have divided the characters to different classes like the common classes are if the character is ‘b’ it belongs to small alphabet class and if it is ‘6’ it belongs to digit class and each class is denoted by a unique code.

Followingis character classes list and their codes representation:

Characters	Character Class	Character Class Code
a-z	Small Alphabet	301
A-Z	Capital Alphabet	302
0-9	Digit	303
. , ; ‘ “	Punctuation	304
+, -, *, /, ^, &, <, >, =	Operator	305
Everything else	Special	306

Sample Program Run 1:

Enter character = P

Output:

Character Class is: Capital Alphabet

Character Class Code is: 302

Sample Program Run 2:

Enter character = P

Output:

Character Class is: Capital Alphabet

Character Class Code is: 302

Q.No.2:

A box of cookies can hold 24 cookies, and a container can hold 75 boxes of cookies. Write a program that prompts the user to enter the total number of cookies, the number of cookies in a box, and the number of cookie boxes in a container. The program then outputs the number of boxes and the number of containers to ship the cookies. Note that each box must contain the specified number of cookies, and each container must contain the specified number of boxes. If the last box of cookies contains less than the number of specified cookies, you can discard it and output the number of leftover cookies. Similarly, if the last container contains less than the number of specified boxes, you can discard it and output the number of leftover boxes

Q.No.3:

You have to create a C++ program which takes **quiz** of a student to check his/her knowledge about arithmetic, logical and relational operations.

- You are provided with a list of expressions below.
- Store the result of these expressions in variables with appropriate data types.
- Prompt the user these expressions one by one and then take answers as inputs.
- Compare each specific user entered answer with values in the corresponding variables.
- Maintain the score of the student and output it when the quiz ends. For each correct answer the student gets one. Use unary operator for it.

Expressions

Bool type:

I. (!0)

II. ((!1 || !0) && (!(1 && 0))

III. (5 + 4 < 3 && 7 + 3 <= 20)

IV. (‘a’ !=’b’ – 1)

V. ((3 % 2) * 1 == 1 && 5 * (3 % 3) == 0)VI. “Ali” >= “Noor”

VII. “123” >= “456”

Int/Float type:

I. (9 / 3 % 3) + (3 * 4 / 4)

II. ((3 – 2 / 2) * 7 % 7

Char:

I. ‘a’ + (5 % 5) * 4

II. ‘Z’ – 5 + (30 / 6)

Note: Implement the following question using for loop.

Q.No.4:

Write a C++ program that generates a random math quiz for the user. The program should use a **for** loop to present a series of math questions. The user should input answers, and the program should provide feedback on whether each answer is correct or incorrect.

Here are the details:

1. The program should ask the user for the number of math questions to generate.

2. Using a `for` loop, generate the specified number of random math questions. Each question should involve two random integers between 1 and 10 and one random arithmetic operator (+, -, *, or /).

3. Present each question to the user in the following format: "Question X: Y [operator] Z = ?" where X is the question number, Y and Z are the random integers, and [operator] is the random operator.

4. Prompt the user to enter the answer for each question.

5. Check if the user's answer is correct, and provide feedback. If the answer is correct, display "Correct!" for that question. If the answer is incorrect, display "Incorrect. The correct answer is [correct answer]."

6. At the end of the quiz, display the user's score as the number of correct answers out of the total number of questions.

Here's an example of how the program should work:

How many math questions do you want to generate? 5

Question 1: 7 + 4 = ? 11
Correct!

Question 2: 9 * 2 = ? 18
Correct!

Question 3: 1 - 3 = ? 2
Incorrect. The correct answer is -2.

Question 4: 6 / 3 = ? 2
Correct!
Question 5: 10 - 7 = ? 3
Correct!

Your score: **4 out of 5**

This question will test your ability to create random math questions, use a `for` loop to present them to the user, and provide feedback on the user's answers**.**

Q.No.5:

Write a C++ program to check if a given number is a rotation of another number without using arrays or strings.

A number is considered a rotation of another if it can be obtained by rotating the digits of the original number. For example, 123 is a rotation of 231.

Note: Implement the following questions using while / do while loop both.

Q.No.6:

1. Initial Numbers for Fibonacci Sequence:
   - Before we start, we need you to give us two positive numbers that will serve as the starting point for the Fibonacci sequence.

2. Comparison of Initial Numbers:
   - It's important to ensure that the first number you provide is less than or equal to the second number.

3. Desired Fibonacci Sequence Position:
   - Now, we'd like to know which specific position in the Fibonacci sequence you're interested in.

4. Validation of Position:
   - Please make sure that the position you mention is a positive number greater than zero.
   - Also, check that the position you've chosen aligns with the initial numbers you provided.

Q.No.7:

Write a program that uses loops to perform the following steps:

a. Prompt the user to input two integers: firstNum and secondNum (where firstNum must be less than secondNum). Ensure that both values are positive integers and handle any input errors gracefully.

b. Output all odd numbers between firstNum and secondNum, but in reverse order (from highest odd number to the lowest odd number).

c. Calculate and output the sum of all even numbers between firstNum and secondNum. Additionally, provide the count of even numbers in that range.

d. Output the numbers and their squares between 1 and 10. Additionally, calculate and output the product of the squares of these numbers.

e. Calculate and output the sum of the square of the odd numbers between firstNum and secondNum, but only for the prime odd numbers within that range.

Q.No.8:

Suppose that the first number of a sequence is x, in which x is an integer. Define a0 = x;

an+1 = an/2 if an is even; an+1 = 3 * an + 1 if an is odd. Then, there exists an integer k

such that ak = 1.

Write a program that prompts the user to input the value of x. The program output the

integer k such that ak = 1 and the numbers a0, a1, a2, . . . , ak.

(For example, if x = 75, then k = 14, and the numbers a0, a1, a2, . . ., a14, respectively,

are 75, 226, 113, 340, 170, 85, 256, 128, 64, 32, 16, 8, 4, 2, 1.)

Test your program for the following values of x: 75, 111, 678, 732, 873 2048, and 65535.

Note: Implement the following question using continue and break

Q.No.9:

Write a program that prints the numbers from 1 to 100, but for multiples of 3, it should print "Fizz," and for multiples of 5, it should print "Buzz." For numbers that are multiples of both 3 and 5, it should print "FizzBuzz." Use **continue** to skip numbers that are not multiples of 3 or 5 and print the appropriate output for the multiples.

Q.No.10:

Write a C++ program that tells the longest sequence of consecutive even numbers in a number, and it breaks when it encounters an odd number after an even number

Note: Implement the following question using Switch Statement

Q.No.11:

Vending Machine design was an open-ended and vague problem. At first thought, many different images flashed my mind and the problem looked intimidating. I decided to drill down the requirements by asking the Interviewer as many questions as I could. After all the clarification, the expectation was to build software they could store items (track the inventory), accept cash, & dispense items. Additionally, the interviewer wanted the design and code to be extensible, reusable & modular. Use **(Switch case)** command.

HINT: You have to set some drinks and set them some valid values that should not exceed more then 1$ (Avoid Negative values)

Requirements:
Following are the requirements for the above discussed problem:
a. Vending Machine must keep track of the inventory
b. A person should be able to insert cash into the machine & choose an item
c. The Machine should confirm the inserted cash with the price of the selected item
d. The machine must display an error in case of insufficient cash or unavailable item
e. Finally, if all the above steps succeed then the user gets the selected item

Q.No.12:

Write a C++ program that simulates a simple calculator. The program should take two numbers and an operator (+, -, *, /) as input from the user. It should then perform the corresponding operation and display the result. Use a switch statement to implement the calculator. Your program should continuously ask the user for input until they choose to exit. If the user enters an invalid operator, display an error message and ask for input again.

GOOD LUCK 😊